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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	NoMoreTokens
Selected Domain	Domain 1: Aiversecity Future-Ready Workforce
Selected SDG Track (1–17)	SDG 4 (Quality Education, target 4.4) and SDG 8 (Decent Work and Economic Growth, targets 8.2 and 8.5)
Project / Solution Name	TomorrowWorks
Name (Leader)	Lee Yun Loong
Email & Phone (Leader)	yunloong6325@gmail.com / +6018-324 8843
University Affiliation(s)	Multimedia University (MMU), Cyberjaya
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

AI is changing which tasks people do at work faster than career advice can keep up. In Malaysia, a fresh graduate with a CV and a job ad, a junior accounting assistant, an HR manager planning training and a regional skills centre all face the same gap. They cannot see, from dated evidence, which roles are growing, how exposed each role is to generative AI, which skills an ad really asks for, and which to build next. Most career tools either give generic chatbot advice or list jobs without explaining them, and neither shows where its numbers come from.

The evidence exists but is scattered. The Department of Statistics Malaysia (DOSM) publishes quarterly online job offers by occupation group (555,509 offers in Q2 2026). The ILO's 2025 index rates 427 occupations for generative-AI exposure; 112 are exposed and 13 sit in the highest gradient. ESCO, the European skills classification, lists the skills each occupation needs. A person cannot combine these sources alone.

Those affected are fresh graduates choosing a first role and writing CVs without knowing which skills each ad asks for, working adults whose routine tasks are being automated (clerical accounting work is in the ILO's highest exposure gradient), and the employers, universities and career centres that fund reskilling. It matters for decent work (SDG 8) and for skills for employment (SDG 4, target 4.4): training spent on the wrong skills wastes money and time that people who cannot afford a wrong turn do not have.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

TomorrowWorks is a web app that takes a person, above all a fresh graduate, from "who I am today" to "where I can be in the AI era": CV, job ad, skills to build, plan.

1. Your CV: the person pastes or uploads a CV (read in the browser, contact details removed) or describes their work in English, Bahasa Malaysia or Chinese. The app suggests ESCO skills and the person ticks the true ones. Check my CV also lists source-cited habits to fix (no AI, nothing stored).

2. A job ad: the person pastes an ad. The app finds its ESCO skills, sorts them into on the CV, held but not shown, and still to build, and names the closest of nine ESCO roles with its DOSM demand and ILO AI-exposure signals. People already working can rank all nine roles by skill fit + demand trend (DOSM) + AI resilience (ILO), each part with source and date; switching the DOSM quarter or AI scenario changes the ranking.

3. Your plan: an ordered learning plan with provider-stated course time, skills not yet covered, and an optional Malaysian NOSS/SKM route. SDG links come from our blind two-reviewer check.

Team and Region views give HR managers, career advisers and regional skills centres group gaps and SDG totals (no names at region level).

AI only maps text onto known lists. Open-weights BAAI/bge-m3 on RunPod embedded all 13,960 ESCO skills once on a GPU Pod (92.5 s); a Serverless endpoint embeds each person's words. It never writes roles, courses or scores. The trend source is updatable: a new DOSM quarter is one import command, and the server checks DOSM daily for a newer one. Workshop skills used: cloud deployment and model serving (Replit, RunPod). No database: nothing about a person is stored.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Front end: React 19, TypeScript 5.9, Vite 6; pdf.js reads PDF CVs and the CV checks run in the browser.

Back end: a dependency-free Node.js 20 server (server/index.mjs) on Replit Autoscale.

AI: BAAI/bge-m3 (open weights) on RunPod, Job A on a GPU Pod and Job B on a Serverless endpoint (Infinity embedding worker).

Data: committed JSON snapshots from DOSM, ILO, ESCO, UN SDG API, hand-checked course and NOSS records; daily server refresh of GDELT news and OpenDOSM labour-force figures (memory only, snapshot fallback). Updatable trend source: a new DOSM Job Market Insights quarter is one import command, and the server checks DOSM once a day for a newer quarter (it never imports by itself).

Tests: Vitest (140). Repository: GitHub, Apache-2.0. No database.

System block diagram:

Browser (React app: scoring, SDG rules, futures simulator, learning plan, on committed snapshots) → Replit Autoscale Node server (static site, /api/skill-search, /api/cv-skills, /api/news, /api/lfs, /api/dosm-next) → RunPod Serverless: bge-m3 embeds the person's text → server ranks the 13,960 precomputed ESCO skill vectors (made once on a RunPod GPU Pod) → nearest real ESCO skills go back to the browser → the person ticks the true ones.

Official SDC sponsor resources: yes. The RunPod credit (US$25) pays for the GPU Pod and the Serverless endpoint above, and Replit credit (US$20) hosts and publishes the app.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathise: the Domain 1 brief names four target groups, and we built fictional personas for them: Aina (working adult), Hakim (fresh graduate, our main persona), Puan Rosnah (HR manager), Encik Lim

(career adviser) and Encik Faizal (regional skills centre officer). We have not yet interviewed real users; that is planned for Phase 2.

Define and ideate: the problem is “which role, which skills, in what order, and why”, so every number on screen shows its source and date, and no model creates scores.

Prototype and test: a working prototype has been deployed since the start and changed in small commits after each walkthrough. After walkthroughs we made fresh graduates our main audience, with a path from CV to job ad to plan. We check it at 390 px and 1300 px, in English, Bahasa Malaysia and Chinese, with a low-data mode for phones. We audited the text intake for identity bias and published the result.

Deployment: Replit Autoscale; the RunPod key lives in Replit Secrets; if RunPod is slow or down, a searchable ESCO list and saved snapshots keep the app usable.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Target 4.4: more youth and adults with relevant skills for decent jobs (baseline for every skill in the app). Target 8.2: higher productivity through technological upgrading. Target 8.5: full and productive employment and decent work for all. Other targets (for example 8.4, 9.4, 12.6 for a sustainability manager) are shown only where two reviewers agreed the link is relevant.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Today, for the two sample people's best moves, 22 skills need building and only 7 (32%) have a hand-checked course; the other 15 (68%) rely on an AI-matched suggestion nobody has checked (measured 23 Oct 2026 with the app's own rule). For SDG links, AI matching alone proposed 73 links and our two reviewers judged 21 relevant (29%). Context: the ILO finds 112 of 427 occupations exposed to generative AI (ILO Working Paper 140, 2025), and DOSM counted 555,509 online job offers in Q2 2026. Our fictional sample ad asks for 24 skills and the sample CV shows 7 (29%) (measured 24 Oct 2026 with the app's own rules).
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Share (%) of a person's skills-to-build, for their chosen move, that have a hand-checked course route. Secondary: share (%) of shown SDG links that two independent reviewers judge relevant.

Metric Category	Your Response
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Raise hand-checked course coverage for the sample people's best moves from 32% to at least 60% (+28 percentage points), and show only team-reviewed SDG links, with any unreviewed link clearly labelled. Beyond that number, Phase 2 also plans: (1) more roles that graduates apply for; (2) a finer Malaysian demand source and a quarterly DOSM import that a person approves; (3) comparing several job ads at once; (4) interviews with real graduates, career centres and HR teams; (5) more hand-checked courses and reviewed Malay and Chinese skill names; (6) optional accounts, with the person's consent, so a graduate or employee can link their own plan directly to their career centre or employer's Team view.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☒ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☒ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☒ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☒ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☒ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React 19.2.7, TypeScript 5.9.3, Vite 6.4.3, Vitest 5.0.5, pdfjs-dist 5.7.284	Code	MIT; Apache-2.0 (TypeScript, pdf.js)
@fontsource/carlito 5.3.0	Code (web font)	OFL-1.1
BAAI/bge-m3	Model (open weights; text to vectors only)	MIT
ESCO v1.2.1 (European Commission); our role list is an adapted selection, labelled as adapted	Dataset	Free reuse with attribution (Decision 2011/833/EU)
DOSM Job Market Insights Q1 and Q2 2026; ILO Working Paper 140 (2025); OpenDOSM lfs_month; UN SDG API target list; GDELT headline metadata	Dataset	Publishers' terms, cited. OpenDOSM: CC BY 4.0. DOSM: Government Open Data Terms of Use 1.0, with attribution.

Microsoft Research "Working with AI" v1.1; Anthropic Economic Index; O*NET OnLine interests	Dataset	CC BY 4.0; CC-BY; CC BY 4.0
Course and certification records: OpenLearn, CS50x, UN CC:Learn, Coursera catalogue (names, links, stated hours only); JPK/JPA NOSS and SKM lists; job tag interests (Japan MHLW)	Dataset	Provider terms; links and facts only, no course content copied
Career-advice articles cited by the CV checks: Jobstreet Malaysia and Singapore, Jobscan, Santa Clara University Career Center; Wilson & Caliskan 2024 (arXiv)	Dataset (web articles; advice rules written by us, no text copied)	Each publisher's terms; cited with link and date

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☒ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



24/10/26